

BiCUT: Checked Single-Level Reformulations for Bilevel Programs with Integer Leaders

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Abstract

BiCUT solves bilevel linear programs with an integer leader through single-level mixed-integer programs and checks every answer it returns. For a continuous follower it uses the big-M KKT reformulation with complementarity on both the constraint slacks and the variable reduced costs. We show that every feasible point of this model is follower-optimal for any value of the big-M constant, so an incorrect constant can only make the model infeasible or its optimum too high. On 122 generated instances with $M = 1000$, HiGHS solves 119 to optimality, every returned point passes a one-LP follower check, and on the 34 instances small enough to enumerate the leader the KKT optimum equals the enumerated optimum. Sweeping M from 1 to 10^4 never produced a follower-suboptimal answer; at $M = 10$, 70 of 117 runs reported infeasibility and 7 returned a feasible but suboptimal value. For an integer follower, the KKT model of the follower’s LP relaxation returns checked bilevel-feasible points, and an exact loop over leader decisions bounded by the high-point relaxation and warm-started from those points proves optimality on 35 of the 80 smallest optimally solved BOBILib instances within 60 seconds each; all 35 proofs agree with the optimum BOBILib reports, and 49 of its final values equal it.

1 Introduction

A bilevel program optimizes a leader’s decision x subject to the condition that a follower responds optimally:

$$\begin{aligned} \min_{x,y} \quad & d^\top x + e^\top y \\ \text{s.t.} \quad & Cx + Dy \leq h, \quad x \in X \cap \mathbb{Z}^{n_x}, \\ & y \in S(x) = \operatorname{argmin}_{y' \geq 0} \{ c^\top y' : Ay' \leq b + Bx \}. \end{aligned} \tag{1}$$

We consider the optimistic version, in which the leader picks the best y among the follower’s optimal responses. Applications include interdiction, pricing and Stackelberg games [3, 4].

The standard way to solve (1) with an off-the-shelf MILP solver replaces $y \in S(x)$ by the follower’s KKT conditions and linearizes the complementarity products with a constant M [10]. Choosing M is the weak point: bound tightening can produce invalid values [15], and verifying that a given M is valid is as hard as solving the bilevel program [12]. When the follower has integer variables the KKT conditions do not characterize optimality at all, and dedicated branch-and-cut solvers are used instead [5, 8, 16].

This paper makes one claim: a single-level reformulation becomes a dependable tool when each of its answers is checked and its failure modes are known in advance. For an LP follower the failure mode of a bad M is restricted to infeasibility or a too-high objective, never a follower-suboptimal point; for an integer follower a checked relaxation-based answer is a valid incumbent for an exact leader enumeration. Contributions:

1. A statement of the big-M KKT model with both complementarity families and a proof that its feasible points are bilevel feasible for every $M > 0$ (Proposition 1).
2. An evaluation on 122 generated instances: correctness against enumeration, a one-LP answer check, and a sweep over M (Section 5.1).
3. For integer followers, a checked KKT heuristic and an HPR-bounded leader enumeration with an optimality proof, evaluated on BOBILib instances with known optima (Section 5.2).
4. An open-source implementation: Python reference code on HiGHS and SCIP, and a Rust workspace with problem types, MPS/LP writers and a command-line tool.

2 Background and Related Work

KKT and big-M. Replacing the follower LP by its KKT system and encoding complementarity with binary variables goes back to Fortuny-Amat and McCarl [10]. Pineda and Morales [15] show that big-M values from common bound tightening heuristics can cut off the bilevel optimum, and Kleinert et al. [12] prove that verifying a big-M is as hard as the original problem. BilevelJuMP [6] offers KKT, strong-duality and SOS1/indicator encodings in Julia. We add a measurement of how often a given M fails and in which way.

Integer followers. With an integer follower, exact methods include the branch-and-bound algorithm of Moore and Bard [14], branch-and-cut by DeNegre and Ralphs [5] and its implementation in MibS [16], and the branch-and-cut algorithm of Fischetti et al. [8], which separates intersection cuts [1] derived from bilevel-free sets [7, 9]. Kleinert et al. [13] survey MILP techniques for bilevel optimization. The leader enumeration in Section 3.3 is a simple reference method for pure-integer leaders; it is not competitive with these solvers on large instances, and we use it to obtain proven optima with standard MIP calls.

Benchmarks. BOBILib [17] collects bilevel instances in an MPS-plus-auxiliary-file format together with the best known or optimal objective values, which we use as reference.

3 Approach

3.1 Big-M KKT model for an LP follower

Let the follower rows be extended by any finite bounds $y \leq u$, giving $A'y \leq b' + B'x$ with m' rows. The follower LP is optimal at y for a fixed x if and only if there is $\lambda \geq 0$ with reduced cost $r = A'^T \lambda + c \geq 0$, $\lambda_k(b' + B'x - A'y)_k = 0$ for all k , and $y_j r_j = 0$ for all j . With binaries $z \in \{0, 1\}^{m'}$ and $t \in \{0, 1\}^{n_y}$ the model is

$$\begin{aligned}
\min \quad & d^\top x + e^\top y \\
\text{s.t.} \quad & Cx + Dy \leq h, \quad A'y - B'x \leq b', \quad A'^T \lambda + c \geq 0, \\
& \lambda \leq Mz, \quad b' + B'x - A'y \leq M(\mathbf{1} - z), \\
& y \leq Mt, \quad A'^T \lambda + c \leq M(\mathbf{1} - t), \\
& x \in X \cap \mathbb{Z}^{n_x}, \quad y \geq 0, \quad \lambda \geq 0, \quad z, t \text{ binary}.
\end{aligned} \tag{2}$$

Proposition 1. *For every $M > 0$, the (x, y) part of every feasible point of (2) is feasible for (1). Hence the optimum of (2) is either infeasible or at least the bilevel optimum, and it equals the bilevel optimum whenever M bounds λ , the slacks, y and r at some optimal bilevel solution with a matching multiplier.*

Proof. If $z_k = 0$ then $\lambda_k \leq 0$, so $\lambda_k = 0$; if $z_k = 1$ the slack of row k is at most 0 and, by primal feasibility, equals 0. The same argument with t_j gives $y_j r_j = 0$. So (y, λ) satisfies the KKT conditions of the follower LP at x , which are sufficient for optimality; the leader constraints are part of the model. The second statement follows because the model is then a restriction of the bilevel program, and a bounded optimal pair is feasible for it. $\square$

A consequence is that a follower check cannot detect a bad M for this model: the answer is always bilevel feasible and the error, if any, is in the leader objective. We still run the check, one LP per answer, as a guard against numerical tolerance and modelling errors. The check accepts (x, y) if all constraints hold and $c^\top y \leq \varphi(x) + 10^{-6}(1 + |\varphi(x)|)$, where $\varphi(x)$ is the follower LP value at x .

3.2 Integer follower: checked KKT of the LP relaxation

When some follower variables are integer, we apply the KKT model to the follower's LP relaxation and keep the integrality of y in the MILP. This is the KKT method in Section 5.2; it uses indicator constraints in SCIP instead of big-M.

Proposition 2. *If (x, y) satisfies the KKT conditions of the LP relaxation of the follower at x and y is integral, then y is optimal for the integer follower at x .*

Proof. $\varphi_{\text{LP}}(x) \leq \varphi_{\text{IP}}(x) \leq c^\top y = \varphi_{\text{LP}}(x)$. $\square$

The model therefore returns bilevel-feasible points, but it excludes every integer response whose value exceeds the LP value, so it can return a suboptimal objective or report infeasibility. We check each answer by solving the follower MIP at the returned x .

3.3 HPR-bounded leader enumeration

For a pure-integer leader with finite bounds we use Algorithm 1. The high-point relaxation (HPR) is (1) without the condition $y \in S(x)$.

A leader vector $\hat{x}$ is excluded with one binary per direction and bound, $x_i \leq \hat{x}_i - 1$ or $x_i \geq \hat{x}_i + 1$ for at least one i , which is valid for general integers.

Proposition 3. *If every sub-MIP is solved to optimality, Algorithm 1 terminates and returns the optimistic bilevel optimum.*

Proof. $F(\hat{x})$ is the optimistic value at $\hat{x}$, so U is always the value of a bilevel-feasible point (or the given incumbent). The bilevel optimum is at least $\min(U, L)$, since every leader vector outside $\mathcal{E}$ has optimistic value at least the HPR value over that set. The loop stops only when $L \geq U$ or the set is empty. Each iteration removes one of finitely many leader vectors. $\square$

The implementation reports a proof only when the loop stops in this way and no sub-MIP hit its time limit.

Algorithm 1 HPR-bounded leader enumeration

Require: bilevel instance, optional incumbent value U of a checked solution

```
1:  $U \leftarrow U$  or  $+\infty$ ;  $\mathcal{E} \leftarrow \emptyset$ 
2: loop
3:   solve the HPR with the leader vectors in  $\mathcal{E}$  excluded
4:   if infeasible or its value  $L \geq U$  then
5:     return  $U$  (optimal)
6:   end if
7:    $\hat{x} \leftarrow$  leader part of the HPR optimum
8:    $\varphi \leftarrow$  follower MIP value at  $\hat{x}$ 
9:    $F \leftarrow \min\{d^\top \hat{x} + e^\top y : Cx + Dy \leq h, Ay \leq b + B\hat{x}, c^\top y \leq \varphi, y \text{ follower-feasible}\}$ 
10:   $U \leftarrow \min(U, F)$ ;  $\mathcal{E} \leftarrow \mathcal{E} \cup \{\hat{x}\}$ 
11: end loop
```

4 Implementation

The reference implementation is in Python. `kkt_highs.py` builds model (2) in HiGHS [11]; `kkt_eval.py` runs the LP-follower evaluation; `bobilib_eval.py` reads BOBILib MPS and auxiliary files and runs the integer-follower methods in SCIP [2]. The repository also contains a Rust workspace with typed bilevel problem structures, MPS and LP readers and writers, reformulation passes as library code with 718 unit tests, and a command-line tool that reads a bilevel problem in TOML and writes its high-point relaxation as MPS or LP.

5 Evaluation

We ask three questions. **RQ1:** does the KKT model with a fixed M return the bilevel optimum on instances with an LP follower? **RQ2:** how does it fail when M is too small? **RQ3:** on integer-follower instances with known optima, how often do the checked KKT heuristic and the leader enumeration reach the optimum?

Setup. HiGHS 1.15.1 and SCIP 10.0 (PySCIPOpt 6.2.1), Python 3.14, on an Apple M2 shared with other jobs; wall-clock times are therefore indicative. KKT solves use a 60 s limit for RQ1 and 30 s for RQ2; each BOBILib method has 60 s per instance.

5.1 RQ1 and RQ2: LP follower

Instances. 122 instances in four generated families (`generators.py`): 72 knapsack-type instances in which a binary leader removes capacity from an LP knapsack with weakly correlated, strongly correlated or uncorrelated weights ($n = 8$ to 50); 28 dense random instances with integer $x \in [0, 3]^{n_x}$ ($5 \leq n_x \leq 20$, $5 \leq n_y \leq 30$); 12 instances with pairwise follower coupling ($n = 5$ to 25); and 10 target-defence games in which a binary defender lowers attack bounds and an LP attacker maximizes damage ($n = 5$ to 20). For the 34 instances with at most 4096 leader vectors we compute the exact optimum by solving one LP per leader vector.

RQ1. Table 1 shows that with $M = 1000$ HiGHS solves 119 of the 122 instances to optimality, every returned point passes the follower check, and the KKT optimum equals the enumerated optimum on all 34 enumerated instances. Of the three dense instances that are not solved, one is

Table 1: KKT model with $M = 1000$ on 122 generated LP-follower instances. “Checked”: the returned point passes the follower-LP check. “Exact”: instances whose leader was enumerated; “=”: KKT optimum equals the enumerated optimum. Nodes and times are means over solved instances.

Family	n	Optimal	Checked	Exact	=	Nodes	Time (s)
Knapsack-type	72	72	72	18	18	1.0	0.25
Dense random	28	25	25	4	4	438.9	8.55
Pairwise coupling	12	12	12	6	6	0.0	0.01
Target defence	10	10	10	6	6	1.0	0.19
Total	122	119	119	34	34		

Table 2: Outcome of the KKT model for different M on the 117 instances with a reference optimum (the enumerated optimum, or the checked optimum at $M = 10^4$).

M	Correct	Infeasible	Suboptimal	Follower-suboptimal	Time limit
1, 2, 5	0	117	0		0
10	40	70	7		0
50	115	1	0		1
100	116	0	0		1
1000, 10^4	117	0	0		0

reported infeasible for every M in the sweep and two reach the time limit with incumbents that pass the check. The linear relaxation of the KKT model is weak outside the knapsack family: its mean gap to the optimum is 47% on the dense family and larger on the coupling and defence families, while the knapsack family has zero gap because leader and follower share the objective there and the high-point relaxation already attains the bilevel optimum. The dense family accordingly needs branching (mean 439 nodes).

RQ2. Table 2 confirms Proposition 1 in practice: of the 976 sweep runs over all 122 instances, 527 returned a point and all 527 pass the follower check. A too-small M shows up in two ways. Most often the model becomes infeasible (every run at $M \leq 5$, 70 runs at $M = 10$, one knapsack instance at $M = 50$). Less often it returns a bilevel-feasible point with a worse leader value: seven dense instances at $M = 10$, with objective errors between 14% and 83% of the optimum. The second case is the dangerous one, because the answer passes every feasibility check; only a run with a larger M or a lower bound reveals it. For these instances every $M \geq 1000$ was sufficient.

5.2 RQ3: integer follower on BOBILib

Instances. We take the 80 smallest general-bilevel instances, by compressed MPS size, that BOBILib reports as solved to optimality: 50 from the DeNegre `miblp` set, 10 Xu–Wang `bmilplib` instances with a mixed-integer follower, 11 MIPLIB-derived instances (`p0033`, `enigma`, `stein27`, `lseu`), 4 T1 instances and 5 small classic instances. All have an integer leader and integer follower variables. Each method has 60s per instance; an objective equals the BOBILib optimum if it agrees to a relative 10^{-5} , since BOBILib prints six decimals.

Table 3: Integer-follower methods on 80 BOBILib instances (SCIP 10.0, 60 s). “Feasible”: the method returned a bilevel-feasible objective (KKT answers are checked with the follower MIP). “= opt”: equals the BOBILib optimum.

Method	Feasible	= opt	Proved optimal
KKT of the LP-relaxed follower	65	16	–
Leader enumeration (Alg. 1), warm-started	79	49	35

RQ3. Table 3 summarizes the results. The KKT model of the LP-relaxed follower returns a checked bilevel-feasible point on 65 instances, as Proposition 2 predicts, but only 16 of these are optimal, and it reports infeasibility on 8 instances that have a bilevel optimum. Both effects come from the same source: the model admits only integer responses whose value equals the LP relaxation value. The leader enumeration, started from the checked KKT value, proves optimality on 35 instances with a median of 2.5 s and a maximum of 39.7 s, and every proof agrees with the BOBILib optimum. On the 44 instances where it stops at the time limit its incumbent equals the optimum 14 times. No method returned an objective below the BOBILib optimum, which is consistent with every returned answer being bilevel feasible. One instance has an unbounded leader variable and is outside the scope of the enumeration. The unproved instances are mostly the larger DeNegre and Xu–Wang instances, where the HPR bound is weak and many leader vectors must be visited; dedicated solvers such as MibS [16] or the method of Fischetti et al. [8] are the appropriate tools there.

6 Discussion and Limitations

The generated instances are small to medium and come from four families, so the observed range of sufficient M values is specific to them; the propositions, not the sweep, are what carries over to other instances. The follower check certifies bilevel feasibility, not optimality, and Proposition 1 shows it cannot reveal a too-small M in the KKT model; for that, a larger M or an independent lower bound is needed. The leader enumeration needs a pure-integer, bounded leader and its running time grows with the number of leader vectors the HPR bound fails to exclude, so it is a reference method for small and medium instances rather than a replacement for dedicated bilevel solvers. All timings were taken on a shared machine.

7 Conclusion

The big- M KKT model, stated with complementarity on both slacks and reduced costs, always returns follower-optimal points; a wrong M costs feasibility or objective value, which we measured on 122 generated instances. For integer followers, the KKT model of the LP relaxation supplies checked incumbents and an HPR-bounded enumeration over leader decisions turns them into proven optima on the BOBILib instances where it finishes. Both methods, the instances and all result files are released with the paper.

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